

# Mayank Pillai

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## PROFESSIONAL SUMMARY

Artificial Intelligence undergraduate with research experience in LLM evaluation, Text-to-SQL, and multilingual retrieval systems, built during a summer research internship at BITS Pilani, alongside full-stack MERN engineering and strong algorithmic fundamentals (160+ LeetCode). Seeking research opportunities in neural networks and applied AI.

## EDUCATION

### Newton School of Technology, Rishihood University

Sonipat, Haryana

*Bachelor of Technology in Artificial Intelligence; CGPA: 9.449/10.0*

*Aug. 2024 – May 2028*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Web Development, Computer Networks, Machine Learning

## EXPERIENCE

### Visiting Summer Research Intern (VISRI)

May 2026 – July 2026

*BITS Pilani | Intent-Driven Spatial Search Engine*

*Pilani, Rajasthan*

- Built and deployed a fully offline multilingual spatial search system on a single local GPU, converting conversational voice and text into executable PostGIS SQL through an LLM intent router with a 3-layer FAISS/LangChain RAG fallback, served via a Gradio + Folium map interface
- Designed a Hybrid Evaluation Framework pairing AST-based structural scoring (sqlglot, 7 clause-level dimensions) with live-execution precision/recall, addressing known gaps in exact-match Text-to-SQL benchmarks
- Benchmarked 4 open-weight Text-to-SQL models zero-shot; Qwen2.5-Coder-7B reached a 78.0% pass rate vs. 46.0% (SQLCoder-8B) and 32.0% (CodeS-7B), generalizing to an unseen campus map at 71.2% over 80 queries
- Curated 186 benchmark queries spanning 5 difficulty tiers and 5 Indian languages, using them to select SeamlessM4T v2 (0.470 WER) over Whisper-Large (0.632) for fully offline speech-to-English

## PROJECTS

### Codebase Copilot | TypeScript, Next.js, Fastify, PostgreSQL (pgvector), Redis, BullMQ, tree-sitter

Sep. 2026

- Built a full-stack tool that ingests any TypeScript/JavaScript GitHub repository and parses it with tree-sitter into a code graph of symbols, imports, calls, and HTTP routes, powering an architecture map, request tracing, and change-impact analysis
- Implemented hybrid retrieval fusing pgvector (HNSW) semantic search with Postgres full-text search via Reciprocal Rank Fusion and one-hop call-graph expansion, streaming answers over SSE with server-validated file/line citations
- Architected a Next.js + Fastify monorepo with a BullMQ/Redis indexing worker and httpOnly-cookie JWT auth, covered by unit, integration, and Playwright tests in GitHub Actions CI and deployed on Vercel and Render

### Hitbox | Next.js, React, Express.js, MongoDB, zod, Vitest, GitHub Actions

Dec. 2025

- Built a full-stack social platform for gamers to track, review, and curate lists from IGDB's 500K+ game catalogue, with follows, an activity feed, and comment threads, deployed on Vercel and Render
- Designed explainable recommendation, backlog-ranking, and taste-match engines as pure, testable scoring functions using genre affinity, member ratings, and IGDB time-to-beat data
- Hardened the Express API with zod validation, JWT/bcrypt auth, rate limiting, and helmet headers, backed by 200+ Vitest/Supertest tests in CI and an axe audit with zero WCAG 2.1 AA violations

### BookConnect | MongoDB, Express.js, React.js, Node.js, Redux Toolkit, Socket.io, Tailwind CSS

Oct. 2025

- Engineered real-time social networking platform with instant messaging using Socket.io WebSocket connections and event-driven architecture
- Managed global state with Redux Toolkit and built a user discovery feed backed by MongoDB compound indexes and query optimization

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, HTML, CSS

**AI/ML:** PyTorch, Hugging Face Transformers, LangChain, FAISS, Gradio; LLM Evaluation, Text-to-SQL, RAG, ASR

**Frontend:** React.js, Next.js, Tailwind CSS, Redux Toolkit, Context API, React Flow

**Backend:** Node.js, Express.js, Fastify, Socket.io, BullMQ, Prisma ORM, zod, RESTful APIs, JWT Authentication

**Databases:** PostgreSQL (pgvector, PostGIS), MongoDB, Redis, MySQL

**Tools & Platforms:** Git, GitHub Actions, Docker, Vitest, Playwright, Vercel, Render, Postman

**Problem Solving:** 160+ LeetCode problems solved (Arrays, Trees, Dynamic Programming, Graph Theory)